

Most real world encryption schemes do not achieve perfect secrecy.

Because perfect secrecy requires

- Key as long as the message
 - Keys used only once
 - Truly random key generation.
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The one-time pad encryption scheme

$$\mathcal{M} = \mathcal{K} = \mathcal{C} = \{0, 1\}^n,$$

$$\text{Gen}: K \leftarrow \mathcal{K}, \quad \text{Enc}_K(M) = C = M \oplus K,$$

$$\text{Dec}_K(C) = C \oplus K$$

is perfectly secret.

For every distribution on $\mathcal{M}$, every $m \in \{0, 1\}^n$ and every $c \in \{0, 1\}^n$ with $P_M[C=c] > 0$,

$$P_M[M=m \mid C=c] = P_M[M=m]$$

PROOF:

$$P_M[C=c \mid M=m] = P_M[M \oplus K = c \mid M=m]$$

[$\because$ By definition of the OTP scheme, $C = M \oplus K$. Since we condition on $M=m$, the event $C=c$ is exactly $m \oplus K = c$.]

Since XOR bijective, i.e., XOR on $\{0, 1\}^n$ is invertible, we have $m \oplus k = c \iff k = m \oplus c$

$$\Rightarrow P_M[M \oplus K = c \mid M=m] = P_M[K = m \oplus c \mid M=m]$$

By construction, K is chosen uniformly from $\{0,1\}^n$ and is independent of M . Hence, $m \oplus c$ has probability $\frac{1}{2^n}$.

$$\Rightarrow P_K[K = m \oplus c | M = m] = 2^{-n}$$

$$\therefore P_K[C = c | M = m] = 2^{-n} \text{ ----- } \textcircled{1}$$

Next we remove the conditioning on M . By the law of total probability,

$$P_K[C = c] = \sum_{m' \in \mathcal{M}} P_K[C = c | M = m'] P_K[M = m']$$

$$= \sum_{m' \in \mathcal{M}} 2^{-n} P_K[M = m'] = 2^{-n} \sum_{m'} P_K[M = m']$$

Since $\{M = m'\}$ exhausts all possibilities, the probabilities over all message must sum to one. i.e., $\sum_{m'} P_K[M = m'] = 1$

$$\Rightarrow P_K[C = c] = 2^{-n} \text{ ----- } \textcircled{2}$$

Using Bayes' theorem, we have

$$P_K[M = m | C = c] = \frac{P_K[C = c | M = m] P_K[M = m]}{P_K[C = c]}$$

$$= \frac{2^{-n} P_A[M=m]}{2^{-n}}$$

$$\therefore P_A[M=m | C=c] = P_A[M=m]$$

This shows the posterior $P_A[M=m | C=c]$ equals the prior $P_A[M=m] \forall m, c$.

$\therefore$ By definition, one-time pad leaks zero information about M and it is perfectly secure.

Despite being perfectly secret, OTP has serious practical limitations.

- Key length = message length

This makes OTP impractical for

- long messages
- Messages with unpredictable lengths.
- Real-time communications where we can't pre-share long keys.

- Key must be used only once.

The XOR attack:

Say an attacker gets two ciphertexts:

$$- C = m \oplus k$$

$$- C' = m' \oplus k$$

then they compute $C \oplus C' = (m \oplus k) \oplus (m' \oplus k)$
 $= m \oplus m'$

This doesn't reveal the full context of m and m' , but it shows exactly where the two messages differ. i.e., it learns which bits are the same or different.

i.e., If the messages are human readable texts, $m \oplus m'$ leaks statistical patterns. These can be used with frequency analysis or known plaintext attacks to recover m, m' .